

# SIPHUMZE LABITI

SOFTWARE ENGINEERING | AZURE DEVELOPER | AI DEVELOPER | AGENTIC  
AI ENGINEER

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✉ Slabiti@live.com | 📞 (27) 60 398 7539 | LinkedIn: Siphumze Labiti

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**C# .NET Core · Python · FastAPI · Django · Angular · React · Azure · Docker · Kubernetes ·  
Playwright · RabbitMQ · SignalR · Java-Spring Boot · Swagger · SQL Server · PostgreSQL ·  
MongoDB · Azure Cognitive Services · OpenAI APIs** ◇

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## SUMMARY

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A results-driven **Full Stack Developer** with over four years of professional experience in end-to-end web, mobile, and AI-powered application development. Proven expertise across **C# (.NET Core, [ASP.NET](#) MVC, Web API)**, **Python (FastAPI, Django)**, and modern frontend frameworks including **Angular** and **React**. Strong background in enterprise software development, event-driven integrations, and automated testing with **Playwright**.

Architected and delivered production-grade solutions including an **IP management system** with Azure Cognitive Services, an **event-driven inventory sync platform** handling 1M+ daily webhooks using RabbitMQ and Redis, and a **production-ready multi-agent AI system** with 9 specialized microservices built on FastAPI and React. Experienced in containerization with **Docker**, orchestration with **Kubernetes**, and CI/CD automation via **Azure DevOps**. Skilled in stakeholder engagement, client requirement-gathering, technical documentation, and second-line business support. Currently pursuing **Databricks Platform Administrator** certification. Seeking to leverage full-stack expertise, cloud-native development, and business engagement skills to deliver robust, scalable solutions in a corporate development environment

## WORK EXPERIENCE

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### ARMSCOR | Software Development Intern

[March-2021]

- Completed intensive software development internship focused on full-stack web development, gaining hands-on experience with **C#**, **.NET Framework**, **SQL Server**, **JavaScript**, **HTML/CSS**, and **Git version control**
- Assisted senior developers in building and testing internal business applications, including writing SQL queries, debugging existing code, and documenting technical specifications
- Participated in daily stand-ups, code reviews, and sprint planning sessions, gaining practical exposure to Agile/Scrum methodologies in a professional software team environment
- Developed a small internal tool for tracking support tickets using **ASP.NET MVC** and **Bootstrap**, which was adopted by the team for daily use

# ARMSCOR [Software Engineer]

[March 2022] – Present

- Engineered a full-stack IP management solution from the ground up using **C# .NET**, **Angular**, and **SQL Server** to automate the complete lifecycle of patents and trademarks, including classification, renewal tracking, and compliance reporting. Integrated **Azure Cognitive Services** for AI-powered document extraction, automatically populating system fields from uploaded PDFs and scanned filings, which reduced manual data entry by **70%** and eliminated human transcription errors. Containerized the entire application using **Docker** and orchestrated deployment on **Azure Kubernetes Service (AKS)** to ensure horizontal scalability and high availability. Designed and maintained CI/CD pipelines in **Azure DevOps** that automated build validation, integration testing, and production rollouts, reducing manual deployment dependencies and enabling faster, more reliable releases. The system improved IP portfolio accuracy, eliminated missed renewal deadlines entirely, and provided stakeholders with real-time visibility into asset status.
- Architected and delivered a production-grade security clearance tracking solution that automated verification workflows for personnel and suppliers working on classified projects. Built on **C# .NET**, **Angular**, and **SQL Server**, the system enforced strict project eligibility rules by dynamically matching security clearance levels to project requirements, preventing unauthorized access at the assignment stage. Integrated **Azure Cognitive Services** for automated identity document validation, including passport and security ID verification, reducing manual verification effort by **80%** and accelerating onboarding times. Implemented a comprehensive end-to-end testing strategy using **Playwright with C#**, achieving **90%** test coverage and reducing regression test execution from 4 hours to just 15 minutes through parallel cross-browser execution in containerized test environments. Managed non-production environments (development, testing, staging) using **Docker** containers, including dedicated Playwright test containers, with all workloads deployed to Windows Server 2022 IIS. Established CI/CD automation on Azure DevOps that executed the full Playwright test suite on every pull request before merging, catching defects early and maintaining production stability. The system reduced clearance verification errors by **95%** and guaranteed that only properly vetted personnel were assigned to sensitive classified projects.
- Engineered event-driven inventory sync platform using **RabbitMQ** and **Redis**, achieving **99.5%** webhook delivery success rate for 50+ B2B resellers. Implemented **Change Data Capture (CDC)** with **Outbox pattern** to guarantee zero data loss across distributed systems. Built real-time dashboard with **SignalR WebSockets**, supporting 50+ concurrent users with sub-second latency. Designed dead letter queue system with exponential backoff retry, reducing failed webhook handling time by **80%**. Containerized microservices using **Docker** and deployed to cloud environment, ensuring horizontal scalability.
- Provided second-line business support for production systems, troubleshooting application issues raised by internal business users and external stakeholders. Attended client requirement-gathering sessions, documented technical specifications, and participated in change-control board meetings to ensure all development followed governance standards. Prepared and maintained technical documentation and user guides for deployed solutions, reducing recurring support queries by **40%**.
- Designed, built, and deployed a production-ready personal AI system integrating **nine specialized microservices** that operate as autonomous agents: file operations (read/write/transform), third-party API integrations, AI image generation, web research and summarization, conversational memory management, social media publishing, calendar and email automation, performance optimization, and voice interaction. Built the backend with **FastAPI (Python)** for high-concurrency asynchronous processing and the frontend with **React** for a responsive user interface. Engineered an intelligent auto-detection routing layer that analyzes user intent in real time and routes requests to the appropriate agent without requiring manual agent selection—a key differentiator from conventional multi-agent tools. Each microservice runs in its own containerized environment, communicating via REST and message queues, with centralized logging and monitoring to ensure reliability in production-like conditions. The system demonstrates advanced capabilities in microservices orchestration, AI integration, and production-ready software design, handling

complex workflows such as "research a topic, generate an image, and email the results" as a single seamless command. Mentored junior developers and interns on C#/.NET best practices, code reviews, and debugging techniques, contributing to team capability growth.

## PROJECTS

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### Intellectual Property Management System (ARMSCOR)

Played a key role in developing a full-stack IP management system using **C# .NET**, **Angular**, and **SQL Server** to automate patent/trademark lifecycle tracking and classification. Integrated Azure Cognitive Services for AI-powered document extraction, reducing manual data entry by 70%. Containerized the application using **Docker** and deployed to **Azure Kubernetes Service (AKS)**. Set up and maintained CI/CD pipeline using **Azure DevOps** to automate builds and deployments, ensuring consistency and reducing manual dependency. This project improved IP portfolio accuracy, eliminated missed renewal deadlines, and enabled faster, reliable production updates.

### Vetting System (ARMSCOR)

Involved in the development of a personnel and supplier vetting solution using **C# .NET**, **Angular**, and **SQL Server** to automate security clearance tracking and project eligibility enforcement. Integrated Azure Cognitive Services for automated identity document validation, reducing manual verification by 80%. Built assignment rules restricting project access only to suppliers and personnel with matching security clearances. Implemented Playwright with C# for end-to-end automation testing, achieving 90% coverage and cutting regression test time from 4 hours to 15 minutes via parallel cross-browser execution. Containerized non-production environments with **Docker** (including Playwright test containers) and deployed to Windows Server 2022 IIS. Established **CI/CD automation** via Azure DevOps running Playwright test suites on every pull request. The system reduced clearance verification errors by 95% and enforced compliant assignment of vetted personnel to classified projects.

### Agent Hub - Production-Ready Multi-Agent System (Personal)

Designed and developed a comprehensive multi-agent AI system integrating 9 specialized microservices for file operations, API integrations, image generation, research capabilities, memory management, social media, calendar/email, performance optimization, and voice interactions. Built with **FastAPI (Python)** backend and **React frontend**, featuring intelligent auto-detection routing that eliminates manual agent selection. **Containerized** with **Docker** and deployed with monitoring and logging for production-like reliability. Implemented financial workflows including automated billing, late fee calculation, and payroll export (CSV/PDF), demonstrating experience in invoicing and financial processing environments. Integrated Claude AI and OpenAI APIs for intelligent agent routing, code generation, and natural language task decomposition.

## **Fleet Management System - Production Ready | React, Django REST API, Bootstrap**

Developed a full-stack fleet management application enabling companies to manage vehicle fleets, employee bookings, maintenance tracking, and analytics. Implemented role-based access control (Employee, Manager, Fleet Admin, Super Admin) with **JWT** authentication. Features include real-time car status tracking, trip check-in/checkout with odometer and fuel monitoring, automated late fee calculation, maintenance scheduling, and interactive dashboards with reporting analytics. Built responsive UI with glass-morphism design, custom toast notifications, and FullCalendar integration. Built with **Django REST API (Python)** and **React**.

## Inventory Sync Hub - B2B Webhook Platform | .NET 8, C#, React, TypeScript, SignalR, RabbitMQ, Redis, SQL Server, Docker

Built an event-driven webhook platform enabling real-time data synchronization across 50+ resellers through a unified integration layer. Implemented Change Data Capture (CDC) with **Outbox pattern** guaranteeing zero data loss and reliable message delivery. Engineered message queuing infrastructure processing 1M+ webhooks daily using **RabbitMQ** with exponential backoff retry strategy and dead letter queue for failed delivery handling. Delivered real-time inventory updates via webhooks with **SignalR WebSockets** supporting 50+ concurrent users for live dashboard monitoring. Developed self-service reseller portal with API key management, webhook testing tools, and subscription configuration. Containerized all **microservices** with **Docker** for consistent deployment across environments.

- Results: 99.5% webhook delivery success rate with automatic retry logic, sub-second latency for inventory sync across 50+ resellers, zero data loss through Outbox pattern implementation, complete monitoring with dead letter queue and analytics dashboard.

## BioClock - Enterprise Biometric Time Tracking System | FastAPI, React, Face Recognition (Facenet-PyTorch), Postgres, Docker, JWT, RBAC

Built a production-ready biometric time tracking system for manufacturing environments, replacing manual paper-based attendance with automated face recognition technology. Engineered 80+ RESTful API endpoints with role-based access control supporting 6 distinct user roles (Worker, Manager, HR, Safety Officer, IT Admin, Supervisor). Implemented face recognition authentication using Facenet-PyTorch (Python), achieving 93% verification accuracy with real-time camera integration for clock-in/out. Designed comprehensive workforce management features including attendance tracking, shift swap requests, incident reporting with manager notes, time-off requests, payroll export (CSV/PDF), and AI-powered absence predictions based on attendance patterns. Containerized full-stack application with Docker featuring multi-container orchestration, health checks, and graceful recovery.

- Results: Eliminated buddy punching fraud, reduced payroll processing errors by 70%, and provided management with real-time floor visibility and automated compliance reporting.

## EDUCATION

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### **Nelson Mandela University (2020-2021)**

#### **BTech Information Technology | NQF7**

- Java, C# Design Patterns, Data Science and Explainable AI

### **National Diploma (With Distinction) | NQF6**

- Majors in Software Development and Information Systems
- Cisco CCNA: Introduction to Networks
- Cisco IT Essentials

### **Certifications**

- **Microsoft Certified: Azure Developer Associate (2025)**

Compute Solutions, Azure Storage, Implement Azure Security, Monitor, Troubleshoot & Optimize, Connect to & Consume Services

- **Databricks Academy Accreditation - Platform Administrator (2026)**

- Identity & Access Management, Unity Catalog Governance, Compute Administration, Workspace Configuration, Automation & CI/CD, Monitoring & Optimization, Cloud-Specific

## SKILLS

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### TECH STACK

#### Languages

C#, JavaScript/TypeScript, Python, Java, SpringBoot Framework, Kotlin, PHP, HTML5, XML, SQL

#### Frontend

Angular, React, React Native, TypeScript, JavaScript, HTML5, CSS3, Razor, Bootstrap, **Tailwind CSS**

#### Backend Frameworks

ASP.NET Core, ASP.NET MVC, Web API, FastAPI, Django, Spring Boot, RESTful APIs, GraphQL

#### Databases & Data Platforms

SQL Server, MySQL, PostgreSQL, NoSQL (MongoDB), **Redis** (Cache/Message Broker), **Databricks**

#### Messaging & Integration

**RabbitMQ**, **SignalR**, **Webhooks**, **Change Data Capture (CDC)**, **Outbox Pattern**

#### AI & Machine Learning

**Azure Cognitive Services** (Document Intelligence, Vision), **Facenet-PyTorch** (Face Recognition), **OpenAI APIs**, **LLM Integration**

#### Testing

**Playwright** (E2E Automation), **xUnit/NUnit** (Unit Testing), **PyTest** (Python Testing), Integration Testing, Regression Testing

#### Cloud & DevOps

**Microsoft Azure** (App Services, AKS, Functions, Storage, Cognitive Services), **Azure DevOps** (CI/CD, Pipelines, Repos), **Docker** (Containerization), **Kubernetes** (AKS), **GitHub Actions**, **Git** (Version Control)

## **Architecture & Design Patterns**

**Microservices**, **Event-Driven Architecture**, **RESTful APIs**, **Message Queuing**, **Containerization**, **CI/CD**, **Agile (Scrum/Kanban)** , **SDLC**, **Design Patterns** (Repository, Outbox, Factory, Dependency Injection)

## **Professional Skills**

Stakeholder Engagement, Client Requirement Gathering, Technical Documentation, **Second-Line Business Support**, **Change Management**, Code Reviews, Sprint Planning, **Production Support & Troubleshooting**